where k is a normalization constant that is often incorporated to ensure that $C_{\boldsymbol{ab}}(0) = 1$.

Using the notation developed earlier for the propagator, this has the alternative formulation

$$C_{\boldsymbol{ab}}(t) = k\mathrm{Av}_{H=\mathrm{E}}\left(\boldsymbol{b}(\boldsymbol{z}) \cdot (\mathrm{e}^{t\mathcal{L}}\boldsymbol{a})(\boldsymbol{z})\right).$$

If the system is ergodic, we could write this as a temporal average with respect to any trajectory initiated on the $H = \mathrm{E}$ energy surface:

$$C_{\boldsymbol{a},\boldsymbol{b}}(t) = k \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^\tau \boldsymbol{a}(\boldsymbol{z}(s+t)) \cdot \boldsymbol{b}(\boldsymbol{z}(s))\mathrm{d}s.$$

The *velocity autocorrelation function* is often written in the shorthand notation

$$C_{\boldsymbol{v},\boldsymbol{v}}(t) \stackrel{\text{def}}{=} \frac{\langle \boldsymbol{v}(t) \cdot \boldsymbol{v}(0)\rangle}{\langle \boldsymbol{v}(0) \cdot \boldsymbol{v}(0)\rangle},$$

where the averaging may be assumed to be against the microcanonical measure, if otherwise unspecified.

Now let $\mathcal{A}$ and $\mathcal{B}$ be subsets of configuration space (i.e. subsets in the position space) of nonzero measure, and define

$$T_{\mathcal{A}\mathcal{B}}(\tau) \stackrel{\text{def}}{=} C_{\mathbf{1}_{\mathcal{B}},\mathbf{1}_{\mathcal{A}}}(\tau),$$

where $\chi_{\mathcal{S}}$ is the *characteristic* or *indicator* function of the set $\mathcal{S}$ defined by

$$\mathbf{1}_{\mathcal{S}}(\boldsymbol{z}) = \begin{cases} 1, \boldsymbol{z} \in \mathcal{S} \\ 0, \boldsymbol{z} \in \bar{\mathcal{S}} \end{cases}.$$

We may understand $T_{\mathcal{A}\mathcal{B}}$ as quantifying the number of trajectories which, started from set $\mathcal{A}$ at time $t = 0$, transition to set $\mathcal{B}$ at time τ. Dividing $T_{\mathcal{A}\mathcal{B}}$ by the measure of the set $\mathcal{A}$, we obtain the probability for a trajectory to make this transition in the given time, which we may write as

$$\Pr\left(\mathcal{A} \to_\tau \mathcal{B}\right) = \frac{\langle \mathbf{1}_{\mathcal{B}}(\boldsymbol{z}(\tau))\mathbf{1}_{\mathcal{A}}(\boldsymbol{z}(0))\rangle}{\langle \mathbf{1}_{\mathcal{A}}(\boldsymbol{z}(0))\rangle},$$

where the averaging is assumed to be with respect to the microcanonical distribution at given energy E.